

(a) Information hierarchy across distinct protocols

schematic — magnitudes are not cross-protocol benchmarks

| Protocol                                                          | Evidence representation                            | Spatial specificity                          | Categorical ambiguity                       | Hard reconstruction outcome                                                                 |
|-------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------|
| Phase 1<br>probability<br><div>ORACLE</div>                       | $P(\text{label}9)$<br>3-D voxel field              | voxel-resolved<br>class-targeted             | low by construction<br>truth-aligned target | 12/12 paired gains<br>IoU <sub>9</sub> 0.031 → 0.810                                        |
| Phase 2<br>property<br><div>ORACLE</div>                          | density + susceptibility<br>3-D property fields    | voxel-resolved<br>all-class mapping          | codebook-dependent<br>contrast sensitive    | 12/12 paired gains<br>IoU <sub>9</sub> 0.031 → 0.481                                        |
| Stage 7<br>structured seismic<br><div>TRUTH-BLIND</div>           | full $x$ - $y$ -time<br>seismic volume             | depth-localized,<br>band-limited response    | petrophysical +<br>wavelet non-uniqueness   | correct arm ranks first 3/3<br>hidden IoU 0.914–0.987                                       |
| Stage 9A<br>unrestricted Flow + seismic<br><div>TRUTH-BLIND</div> | full $x$ - $y$ -time seismic<br>+ frozen Flow pool | global likelihood;<br>weak target enrichment | prior support +<br>likelihood alignment     | 3072 unique hard models<br>0 full-support candidates<br>SUPPORT=FAIL<br>DISCRIMINATION=FAIL |

Oracle fields test controllability; structured and unrestricted seismic rows test different truth-blind inference spaces.